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Limits to northward drift of the Paleocene Cantwell Formation, central Alaska

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ABSTRACT

Volcanic rocks of the Paleocene Cantwell Formation in central Alaska apparently originated at a paleolatitude of 83°N (α_{95} = 9.7°), as indicated by paleomagnetic results. When compared with the Paleocene pole for the North American craton, the 95% confidence limits of the results suggest that terranes north of the Denali fault have moved no more than 550 km northward relative to the North American craton since Paleocene time.

INTRODUCTION

The Cantwell Formation in the central Alaska Range consists of continental sedimentary rocks and calc-alkalic volcanic rocks that were deposited during the Paleocene. We made a paleomagnetic study of these volcanic rocks in order to establish the early Tertiary latitude of the basin in which the Cantwell Formation was deposited. Results of the study constrain plate-tectonic models that propose large-scale northward movements of Alaskan crustal fragments during the Tertiary.

Alaska has been described as an assemblage of crustal fragments (Fig. 1) that have accreted to the continental margin (Coney and others, 1980; Jones and others, 1982a).

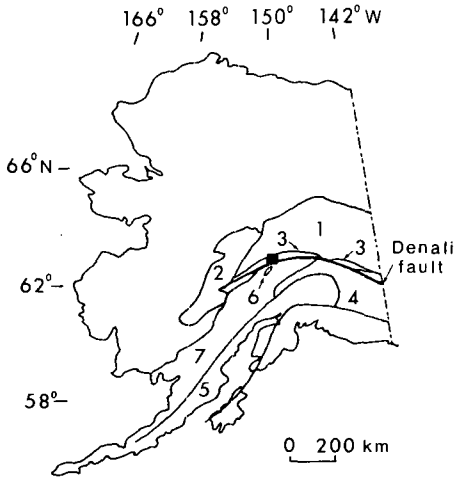


Figure 1. Index map of Alaska showing area of this study (square) and selected tectonostratigraphic terranes according to Coney and others (1980): 1, Yukon-Tanana; 2, Nixon Fork; 3, Pingston and McKinley, undivided; 4, Wrangellia; 5, Peninsular; 6, Chulitna; 7, deformed upper Mesozoic (Upper Jurassic and Lower Cretaceous) flysch.

Paleomagnetic studies, which can detect differences in paleolatitude between the craton and adjacent terranes, provide a way to test the accretion hypothesis. The available paleomagnetic data from Mesozoic rocks of Wrangellia (Hillhouse, 1977; Jones and others, 1977) and the Alaska Peninsula (Packer and Stone, 1974; Stone and Packer, 1979) favor the hypothesis that parts of southern Alaska have moved thousands of kilometres north relative to the North American craton. Not only is the timing of northward drift critical to Alaskan geological studies, but it also bears on the reconstruction of early plate motions in the Pacific. Paleomagnetic results from lower Tertiary rocks of the Cantwell Formation place firm limits on Tertiary movements of terranes located north of the Denali fault (McKinley strand); by inferences drawn from the distribution of lower Tertiary igneous rocks, the results constrain the timing of northward movement of the Peninsular terrane and Wrangellia.

STRATIGRAPHY AND PALEOMAGNETIC SAMPLING

Lower Tertiary rocks of the Cantwell Formation occur in the vicinity of Mount McKinley National Park in the central Alaska Range (Fig. 2). The now-deformed basin in which the Cantwell was deposited lies north of the McKinley strand of the Denali fault. As described by Brooks (1911), the Cantwell Formation consists of conglomerates, coarse quartz-rich sandstones, and volcanic beds, which overlie highly deformed upper Paleozoic, Upper Triassic, and upper Mesozoic rocks (Reed, 1961; Hickman, 1974). The highly deformed basement rocks have been described as parts of the Pingston and McKinley terranes by Jones and others (1982a). The upper part of the Cantwell consists of a thick sequence of andesite, rhyolite, and basalt flows, felsic pyroclastic rocks, and related intrusive rocks. Gilbert and others (1976) described this volcanic assemblage as the Teklanika Formation, and they stratigraphically restricted the name Cantwell Formation to the predominantly sedimentary rocks in the lower part of the Cantwell as originally defined by Brooks (1911). We have retained Brooks's (1911) original definition of the Cantwell because the restriction of the name violates Article 14b of the Code of Stratigraphic Nomenclature (American Commission on Stratigraphic Nomenclature, 1974). Only Quaternary deposits overlie the Cantwell Formation.

The six whole-rock isotopic ages that have been obtained from volcanic rock of the Cantwell Formation by the potassium-argon method range from 41.8 to 59.3 m.y., and plagioclase separated from an andesite flow and a quartz-diorite plug yielded ages of 60.6 and 57.2 m.y., respectively (Gilbert and others, 1976; Hickman, 1974). These ages are considered to be minimum ages of the volcanic rocks because partial argon loss due to incipient alteration is suspected. Plant fossils from the lower part of the Cantwell Formation suggest a Paleocene age (Wolfe and Wahrhaftig, 1970; Wolfe, 1972). On the basis of these data and in view of the minimum isotopic ages, the overall age of the Cantwell Formation, including the Teklanika Formation of Gilbert and others (1976) is considered Paleocene.

We collected paleomagnetic samples from volcanic rocks of the Teklanika at 19 sites, each representing a separate lava flow or cooling unit that presumably preserves an instantaneous reading of the ancient

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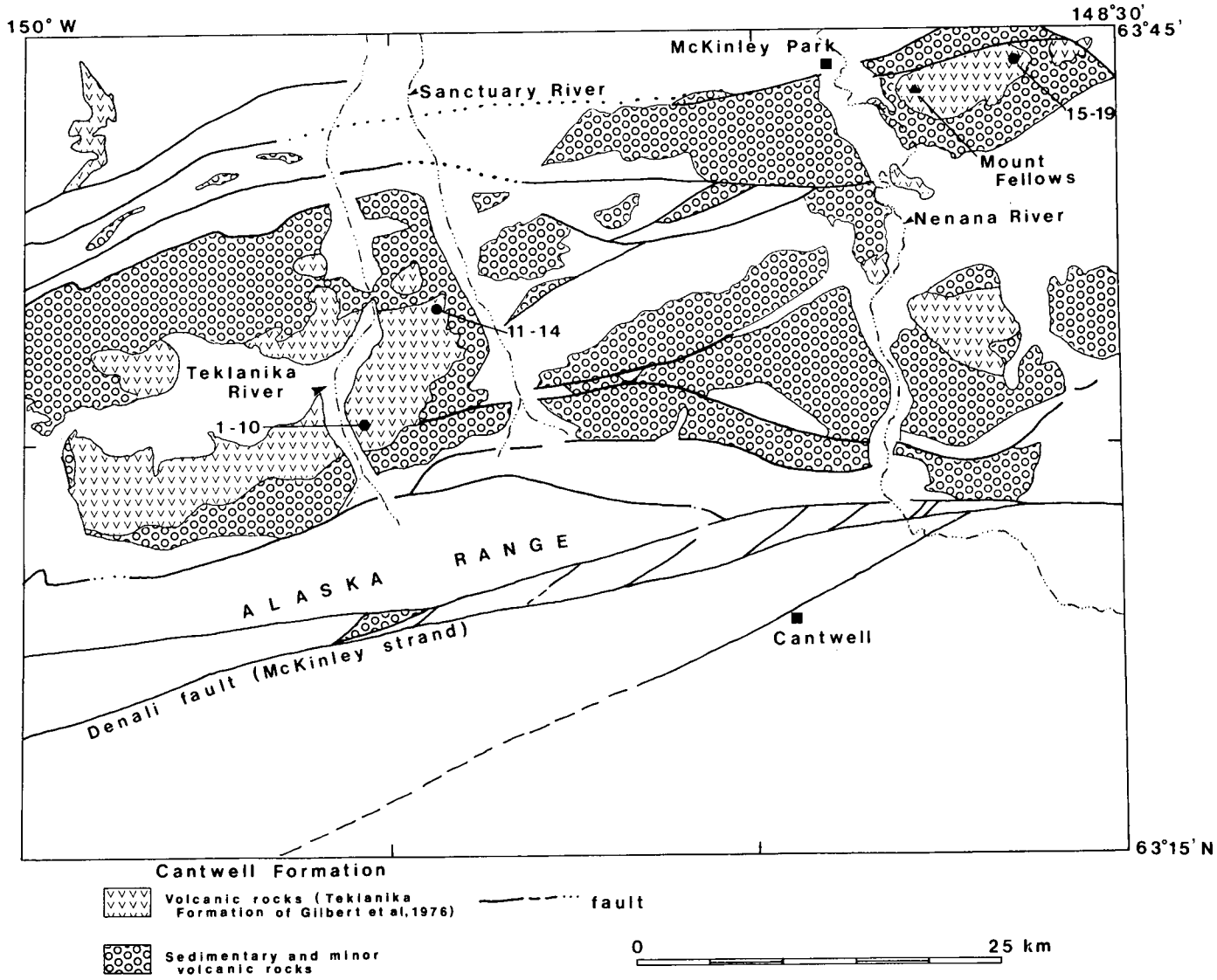


Figure 2. Location of paleomagnetic sites (black dots) in Teklanika Formation of Gilbert and others (1976). 1–10, Type section; 11–14, Sanctuary River; 15–19, Mount Fellows. Geology from Jones and others (1982b).

geomagnetic field direction. Each site is represented by eight cores, oriented by means of a magnetic compass. We corrected the magnetic azimuths for local magnetic anomalies by backsighting to several landmarks. Sampling was restricted to outcrops that have excellent exposure and unequivocal structural control. Our sites are concentrated in three areas spread over a distance of 50 km (Fig. 2).

The westernmost group of sites (1–10) spans 2,000 m of the Teklanika type section that was established by Gilbert and others (1976) near the Teklanika River. The type section is a northeast-striking homocline that generally dips from 40° to 60° to the Northwest. Sites 1 through 5 were taken in vesicular basalt flows (unit B of Gilbert and others, 1976), 2 to 3 m thick. According to Nye (1978), the basalt consists of felted plagioclase laths and subordinate phenocrysts of augite and hypersthene. Sites 6 through 10 are in the upper part of the section (units D and E of Gilbert and others, 1976), which consists of crystal-rich welded rhyolite tuff, clayey tuffaceous beds, and amygdaloidal olivine-augite basalt flows.

The remaining two groups of specimens were collected from basalts exposed in a short stratigraphic section west of the Sanctuary River and from isolated outcrops of basalt on Mount Fellows (Fig. 2). Sites 11 through 14 were collected from the hinge of a gently plunging syncline, in roughly the same stratigraphic position as sites 1 through 5. The stratigraphic positions of the flows sampled on Mount Fellows (sites 15–19) are not known relative to the more westerly sites, because the Teklanika exhibits major lateral facies changes and exposures of the formation are not continuous.

PALEOMAGNETIC ANALYSIS AND RESULTS

Despite evidence of localized hydrothermal alteration, the volcanic rocks yielded paleomagnetic results of high quality, as is commonly observed in fresh, unaltered basalts. With the exception of one site, the natural remanent magnetizations had well-clustered directions, relatively free of secondary magnetizations. The one exception, site 17, exhibited unusually intense, scattered directions, which is a symptom of magnetic contamination caused by lightning. One specimen from each site was progressively demagnetized in alternating magnetic fields up to 100 mT, and the

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directional stability of the magnetization was analyzed from orthogonal projections of the results. Magnetic directions stabilized after treatment in applied fields of 15 to 50 mT, so the remaining specimens were treated in this range of alternating fields. The treatment, which removed minor magnetic contamination due to the recent geomagnetic field (viscous remanent magnetization) or sample transport and preparation, decreased the angular dispersion of the original directions in all cases except site 17. The contamination due to lightning could not be removed, so the result from site 17 was omitted from Table 1.

We selected sample localities so as to maximize the variety of structural attitudes, thereby providing a test of whether the magnetization predates the major deformation of the region. The fold test is a comparison of the directional dispersion before and after the corrections for tilt of the bedding are applied. If the tilt corrections decrease the directional dispersion, then the magnetization predates the deformation. Magnetic field directions from the Teklanika conclusively pass this test (Fig. 3).

The means for tilt-corrected sites, which represent a great thickness of the formation, all give magnetic directions with steep upward inclinations, indicating that the Cantwell Formation was deposited near the north geographic pole during a long period of reversed polarity. Because the Paleocene is dominated by periods of reversed polarity (Ness and others, 1980), this observation is consistent with the assigned age of the Cantwell. Moreover, we see no reason to doubt that the magnetization was acquired during original cooling of the lavas, because subaerial basalts are typically found to be extremely stable magnetic recorders, the fold test is strongly positive, and the magnetization exhibits excellent directional stability.

The paleolatitude of the site of deposition can be inferred from paleomagnetic inclinations, if the inclination is accurately known relative to the true paleohorizontal plane and, when averaged, the data represent the ancient axial dipole field. Although the volcanic flows of the Teklanika are well bedded in places, it is conceivable that the flows were emplaced with an initial dip, especially near volcanic vents. In view of the broad geographic distribution of sites, the dramatic decrease of the directional dispersion following application of highly varied tilt corrections, and the presence of lacustrine interbeds, we believe that the flow bedding was horizontal, and initial dip does not pose a problem for our study. The amount of time spanned by the part of the Teklanika we sampled is uncertain, although inferences drawn from the stratigraphy suggest that the time interval is adequate to yield a mean paleomagnetic pole equivalent to the ancient geographic pole. The stratigraphic section is very thick and exhibits highly varied petrology and geochemistry (Gilbert and others, 1976). Deposition of the volcanic rocks during a single reversed polarity interval does not necessarily imply a brief depositional period, because the Paleocene is dominated by reversed chrons lasting as long as 2.5 m.y. (Ness and others, 1980).

In structurally complex areas such as the Alaska Range, the calculation of mean paleomagnetic poles is complicated by uncertainties in declination due to local rotations. The uncertainties usually arise from folding about nonhorizontal axes, from oblique slip on faults, or from block rotations in shear zones. The simple bedding correction used in our study will not remove errors in declination created by these sources. On the other hand, the exist-

TABLE 1. PALEOMAGNETIC DATA FROM PALEOCENE VOLCANIC ROCKS, CENTRAL ALASKA

Site	N	Treatment (mT)	R	k	α_{95}	I	D	I_c	D_c	Θ	ϕ
1	8	20	7.8346	42	8.6	-46.2	339.3	-76.7	111.6	61.6	154.0
2	8	30	7.9883	596	2.3	-43.7	337.6	-79.4	107.1	62.1	164.5
3	7	20	6.9808	313	3.4	-41.9	325.0	-83.7	169.0	75.6	200.9
4	8	30	7.9887	622	2.2	-47.1	318.8	-77.0	177.0	87.9	174.3
5	8	30	7.9817	382	2.8	-44.2	319.2	-79.4	184.7	83.7	225.7
6	8	30	7.7825	32	9.9	-71.8	343.9	-74.2	166.2	82.9	101.4
7	8	30	7.9458	129	4.9	-46.5	21.1	-65.8	67.1	33.4	163.0
8	8	20	7.9745	274	3.3	-52.7	359.8	-80.8	60.2	51.5	184.9
9	8	20	7.9739	268	3.4	-51.8	350.1	-84.9	22.9	54.0	203.8
10	8	30	7.9804	357	2.9	-57.9	339.0	-86.2	222.4	68.5	224.3
11	8	20	7.9942	1216	1.6	-57.8	251.3	-85.7	320.0	56.7	220.5
12	8	20	7.9852	474	2.6	-54.7	256.8	-81.6	308.4	51.3	231.4
13	8	20	7.9906	743	2.0	-51.0	257.3	-78.7	294.0	49.9	242.2
14	8	40	7.9628	188	4.1	-58.6	248.1	-87.6	326.7	59.4	215.8
15	8	30	7.9847	459	2.6	-68.6	227.6	-80.9	153.8	77.2	173.7
16	8	40	7.9843	445	2.6	-58.7	200.9	-65.8	161.9	71.4	72.2
18	8	20	7.9980	3427	1.0	-72.7	249.2	-65.9	168.1	73.2	59.8
19	8	50	7.9258	94	5.7	-66.1	144.1	-83.3	10.6	50.6	207.4

Mean paleomagnetic pole of 18 flows: 70.0°N., 194.6°E., α_{95} = 10.0°

Note:
N: number of specimens; treatment: alternating field (milliTesla); R: resultant of N unit vectors; k: Fisher (1953) precision parameter; α_{95}: 95% confidence limit for directions; I,D,I_c,D_c: inclination and declination for in situ and tilt-corrected directions; Θ, and ϕ: north latitude and east longitude, respectively, of virtual north geomagnetic pole.

Site coordinates: sites (1-5), 63.51°N, 210.47°E; sites (6-10), 63.53°N, 210.46°E; sites (11-14), 63.59°N, 210.55°E; sites (15-16), 63.73°N, 211.36°E; sites (17-18), 63.73°N, 211.35°E; site (19), 63.72°N, 211.27°E. See Figure 2 for locations.

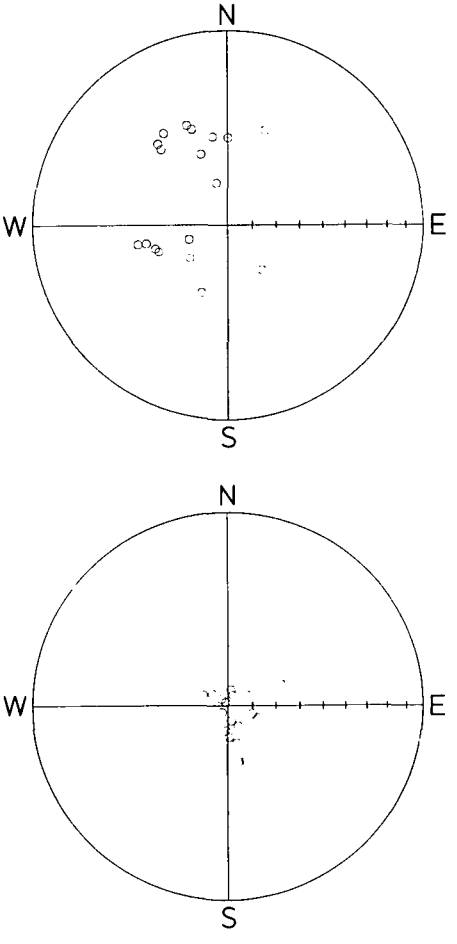


Figure 3. Site mean magnetic directions before (top) and after (bottom) application of corrections for tilt of bedding. Equal-area projection of upper hemisphere.

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ence of complex structures will not introduce errors in magnetic inclination with respect to the bedding planes. Therefore, we prefer to use inclinations rather than mean directions in calculating the paleolatitude of the Cantwell Formation. To do this, we first transformed each site-mean inclination to the corresponding virtual geomagnetic latitude (VGL), assuming a dipole field. The mean paleolatitude was then calculated by Kono's (1980) statistical method, which assumes that the poles corresponding to the VGLs follow the distribution of Fisher (1953). The calculation yielded a mean paleolatitude of 83.0° with a 95% confidence limit of ± 9.7° for the Teklanika Formation. We have also calculated a mean paleolatitude based on the mean paleomagnetic pole (70.0°N, 194.6°E), assuming that declination errors are negligible. This calculation gave a mean paleolatitude of 81.0° (α_{95} = 10°), which is not significantly different from the result derived by Kono's (1980) method.

DISCUSSION

Determining the proper cratonic reference pole for comparison with data from the Teklanika is complicated by age uncertainties and the rapid movement of North America during early Tertiary time. North American paleomagnetic poles of Cretaceous age, but older than 77 m.y. cluster near the Bering Strait (68°N, 186°E, α_{95} = 2.3°; Fig. 4; Mankinen, 1978). After a long stationary period, which ended some time between 77 and 65 m.y. ago, the North American cratonic plate began to move rapidly, probably in conjunction with the Laramide orogeny (Coney, 1978). By 65 m.y. ago the pole had moved 10° northward to 80.5°N, 185.1°E (α_{95} = 5.6°), as is indicated by paleomagnetic results from Paleocene intrusive rocks in Montana (Jacobson and others, 1980). From isotopic dating, we know that the Teklanika is probably older than 60 m.y., and because the formation records reversed geomagnetic polarity, it must be younger than 80 m.y., the end of the Cretaceous long normal chron (Irving and Pullaiah, 1976). Also, plant fossils from the lower part of the Cantwell Formation indicate a Paleocene age. Therefore, the volcanic rocks could have been erupted during the final stages of the stationary period or during the early Tertiary period of rapid apparent polar wander. We find no significant difference between the observed magnetic latitude of the Teklanika (83.0° ± 9.7°) and the expected latitude that is calculated from the Late Cretaceous (80.0° ± 2.3°) or Paleocene (71.8° ± 5.6°) reference poles (Fig. 4), although the mean paleolatitude of the Teklanika is more compatible with the Late Cretaceous reference pole. Taken at face value, the results suggest that the Cantwell Formation was deposited at a latitude higher than would be predicted by the reference poles, and that subsequently these rocks have moved south relative to the craton. The implied amount of relative motion is not significant in the context of the 95% confidence limits.

CONCLUSIONS

The clear implication of the paleomagnetic results is that, within the precision of the method, all terranes lying beneath or north of the present-day position of the Cantwell Formation have undergone no detectable poleward drift relative to the craton since the Paleocene. (The paleomagnetic results do not constrain possible relative east-west motions.) The maximum allowable northward drift, as suggested by the 95% confidence limits of the paleolatitudes, is about 5° of latitude (550 km). This limit applies to the larger terranes such as Yukon-Tanana and Nixon Fork and the smaller terranes such as McKinley and Pingston (Jones and others, 1982a). Whether this limit applies to terranes south of the Denali fault, such as Wrangellia and the Peninsular, depends critically on the relationship of volcanism in the Cantwell Formation with early Tertiary plutons and volcanic rocks that are part of the southern terranes. If these suites of volcanic and plutonic rocks are cogenetic, then our results would indicate also that terranes immediately south of the Denali fault have not drifted north significantly since the Paleocene.

Gilbert and others (1976) noted geochemical similarities between the Teklanika Formation and early Tertiary plutonic rocks from the Alaska–Aleutian Range batholith (Reed and Lanphere, 1973). The

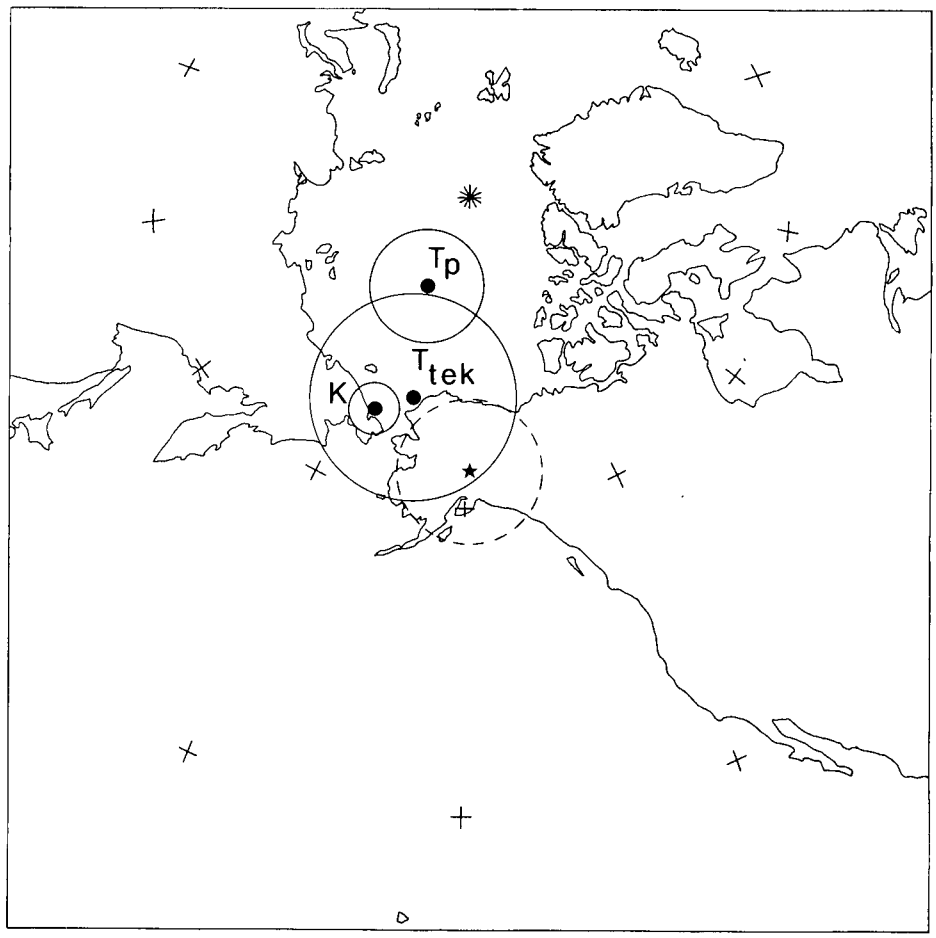


Figure 4. Paleomagnetic poles and 95% confidence circles for T_{tek}, Teklanika Formation (Paleocene) of Gilbert and others (1976) (this report); T_p, Paleocene intrusive rocks from Montana (Jacobson and others, 1980); K, North American craton (Cretaceous; Mankinen, 1978). Star indicates study area. Dashed circle indicates locus of poles corresponding to mean geomagnetic latitude of Teklanika according to Kono's (1980) statistical method. Asterisk indicates present geographic pole.

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plutonic belt, which gives K-Ar ages between 83 and 59 m.y., intrudes the Peninsular terrane from Mount McKinley to Iliamna Lake (Reed and Lanphere, 1973). Therefore, large-scale northward drift of the Peninsular terrane relative to the craton, as indicated by Jurassic paleomagnetic results (Stone and Packer, 1979), would have ceased by Paleocene time if the batholith and the Teklanika are cogenetic. Wrangellia's northward drift would have ceased at the same time, because overlapping flysch deposits indicate that Wrangellia and the Peninsular terrane were amalgamated by the Middle Jurassic.

Our inferences concerning the cessation of the drift of the southern terranes are contradicted by paleomagnetic results from early Tertiary sedimentary rocks on the Alaska Peninsula (Stone and others, 1982). Paleolatitudes determined from four early Tertiary localities in sedimentary rocks on the peninsula are about 30° lower than the paleolatitude of the region estimated from cratonic reference poles. Taken at face value, early Tertiary paleomagnetic data from the Cantwell Formation and the Peninsular terrane require thousands of kilometres of subduction or right-lateral movement along the McKinley strand of the Denali fault. Additional paleomagnetic studies of early Tertiary volcanic rocks in the Talkeetna Mountains and other parts of the Peninsular terrane will test the validity of paleolatitudes inferred from Peninsular sedimentary rocks and should clarify the accretionary history of southern Alaska.

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